

Algebraic Topology I: Homework 2

Exact Sequences, Chain Homotopies, Mapping Cylinders, and Cones

Work in an abelian category $\mathcal{A}$, except where modules are specified. Use homological grading: $d_n^C : C_n \rightarrow C_{n-1}$. Write $Z_n(C) = \ker d_n^C$, $B_n(C) = \operatorname{im} d_{n+1}^C$, and $H_n(C) = \operatorname{coker}(B_n(C) \rightarrow Z_n(C))$. The shift is $C[1]_n = C_{n-1}$ with differential $-d_{n-1}^C$. A chain map is a *quasi-isomorphism* if it induces isomorphisms on all homology objects. You may use Homework 1 and the snake lemma. In $\mathcal{A}$, use morphisms and universal properties; element arguments are permitted for modules. All exercises are required.

Exercise 1. Chain Homotopies

A homotopy from $g : C \rightarrow D$ to $f : C \rightarrow D$ is a family $h_n : C_n \rightarrow D_{n+1}$ satisfying

$$f_n - g_n = d_{n+1}^D h_n + h_{n-1} d_n^C.$$

A complex is *contractible* if its identity is homotopic to zero. A chain map $f : C \rightarrow D$ is a *chain homotopy equivalence* if there is a chain map $g : D \rightarrow C$ with $gf \simeq \operatorname{id}_C$ and $fg \simeq \operatorname{id}_D$.

- Prove that chain homotopy is an equivalence relation on chain maps and is compatible with addition and composition. Thus complexes and chain-homotopy classes of maps form an additive category $K(\mathcal{A})$.
- Prove that homotopic chain maps induce the same maps on homology. Deduce that every chain homotopy equivalence is a quasi-isomorphism, and every contractible complex is acyclic.
- Suppose $i : C \rightarrow D$ and $r : D \rightarrow C$ are chain maps with $ri = \operatorname{id}_C$ and $ir \simeq \operatorname{id}_D$. Construct the induced inverse isomorphisms on homology. Give an example in which neither i nor r is an isomorphism of complexes.

Exercise 2. The Long Exact Sequence in Homology

Let $0 \rightarrow A \xrightarrow{i} B \xrightarrow{\rho} C \rightarrow 0$ be a short exact sequence of complexes, meaning that it is exact in every degree.

- For modules, define $\delta_n : H_n(C) \rightarrow H_{n-1}(A)$ as follows: lift a cycle $c \in C_n$ to $b \in B_n$, and let $a \in A_{n-1}$ satisfy $i(a) = d^B(b)$. Prove existence and uniqueness of a , that a is a cycle, and that sending the class of c to the class of a is well defined and linear.
- Construct δ_n in an arbitrary abelian category using the snake lemma. Specify the diagram and justify every kernel and cokernel identification. Prove exactness at all three types of terms in

$$\cdots \rightarrow H_n(A) \xrightarrow{H_n(i)} H_n(B) \xrightarrow{H_n(\rho)} H_n(C) \xrightarrow{\delta_n} H_{n-1}(A) \rightarrow \cdots.$$

- For a morphism (a, b, c) between two short exact sequences, prove that the square

$$\begin{array}{ccc} H_n(C) & \xrightarrow{\delta_n} & H_{n-1}(A) \\ H_n(c) \downarrow & & \downarrow H_{n-1}(a) \\ H_n(C') & \xrightarrow{\delta'_n} & H_{n-1}(A') \end{array}$$

commutes. Deduce that if two of a, b, c are quasi-isomorphisms, so is the third.

Exercise 3. Degreewise Splittings

Suppose $0 \rightarrow A \xrightarrow{i} B \xrightarrow{\rho} C \rightarrow 0$ splits in each degree. Choose sections $s_n : C_n \rightarrow B_n$ of ρ_n . Identify B_n with $A_n \oplus C_n$ using the unique map whose restrictions to the summands are i_n, s_n .

- (a) Show that the differential has the form

$$d_n^B = \begin{pmatrix} d_n^A & \tau_n \\ 0 & d_n^C \end{pmatrix}, \quad \tau_n : C_n \rightarrow A_{n-1},$$

and that τ is a chain map $C \rightarrow A[1]$. Identify the connecting morphism of Exercise 2 with $H_n(\tau)$, including its sign.

- (b) Replace s_n by $s'_n = s_n + i_n u_n$. Compute τ'_n and exhibit a chain homotopy between τ and τ' . Prove that the short exact sequence splits as a sequence of complexes if and only if τ is nullhomotopic.
- (c) Give a degreewise split short exact sequence of complexes of abelian groups whose connecting homomorphisms all vanish but which does not split as a sequence of complexes. Specify the complexes and maps, and prove both assertions.

Exercise 4. Mapping Cones

For a chain map $f : C \rightarrow D$, define

$$\text{Cone}(f)_n = D_n \oplus C_{n-1}, \quad d_n^{\text{Cone}(f)} = \begin{pmatrix} d_n^D & f_{n-1} \\ 0 & -d_{n-1}^C \end{pmatrix}.$$

For any biproduct, ι_j and ρ_j denote its canonical inclusions and projections.

- (a) Verify that $\text{Cone}(f)$ is a complex and that the structure maps give a degreewise split short exact sequence

$$0 \rightarrow D \xrightarrow{\iota_1} \text{Cone}(f) \xrightarrow{\rho_2} C[1] \rightarrow 0.$$

- (b) Compute the connecting morphism, including its sign, and write the resulting long exact sequence using $H_*(C)$, $H_*(D)$, and $H_*(\text{Cone}(f))$. Prove that f is a quasi-isomorphism if and only if $\text{Cone}(f)$ is acyclic.
- (c) If $f - g = d^D h + h d^C$, construct an isomorphism $\text{Cone}(f) \rightarrow \text{Cone}(g)$ which is the identity on the subcomplex D and on the quotient $C[1]$. Give its inverse and check its sign.
- (d) Construct a contracting homotopy for $\text{Cone}(\text{id}_C)$, and identify $\text{Cone}(0 : C \rightarrow D)$ as a biproduct of complexes.

Exercise 5. Mapping Cylinders

For a chain map $f : C \rightarrow D$, define

$$\text{Cyl}(f)_n = D_n \oplus C_n \oplus C_{n-1}, \quad d_n^{\text{Cyl}(f)} = \begin{pmatrix} d_n^D & 0 & f_{n-1} \\ 0 & d_n^C & -\text{id}_{C_{n-1}} \\ 0 & 0 & -d_{n-1}^C \end{pmatrix}.$$

Let $i : C \rightarrow \text{Cyl}(f)$ and $j : D \rightarrow \text{Cyl}(f)$ be the second and first inclusions, respectively. Define $p : \text{Cyl}(f) \rightarrow D$ by its three restrictions $\text{id}_D, f, 0$.

- (a) Verify that the differential squares to zero, that i, j, p are chain maps, and that

$$\begin{array}{ccc} C & \xrightarrow{i} & \text{Cyl}(f) \\ & \searrow f & \downarrow p \\ & & D \end{array}$$

commutes. Prove that i is a degreewise split monomorphism and $pj = \text{id}_D$.

- (b) Construct $H_n : \text{Cyl}(f)_n \rightarrow \text{Cyl}(f)_{n+1}$ with $\text{id}_{\text{Cyl}(f)} - jp = dH + Hd$ and $Hj = 0$. Deduce that every chain map factors as a degreewise split monomorphism followed by a chain homotopy equivalence.
- (c) Define $q : \text{Cyl}(f) \rightarrow \text{Cone}(f)$ by its two components ρ_1, ρ_3 . Prove that

$$0 \rightarrow C \xrightarrow{i} \text{Cyl}(f) \xrightarrow{q} \text{Cone}(f) \rightarrow 0$$

is degreewise split exact. Identify its long exact sequence with the cone sequence of Exercise 4, using $H_*(p)$ to identify $H_*(\text{Cyl}(f))$ with $H_*(D)$. Determine the sign of the connecting map $H_n(\text{Cone}(f)) \rightarrow H_{n-1}(C)$.

Exercise 6. Acyclic Versus Contractible

Let $f : C \rightarrow D$ be a chain map.

- (a) Prove that $f : C \rightarrow D$ is a chain homotopy equivalence if and only if $\text{Cone}(f)$ is contractible.
- (b) Over a field, prove that every acyclic complex is contractible, without assuming boundedness. Deduce that every quasi-isomorphism of complexes of vector spaces is a chain homotopy equivalence.
- (c) Let P be $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees 1, 0, and let Q have $\mathbb{Z}/2$ in degree 0 and zero elsewhere. Let $f : P \rightarrow Q$ be reduction modulo 2 in degree 0. Prove that f is a quasi-isomorphism but is not a chain homotopy equivalence. Describe $\text{Cone}(f)$ and show directly that it is acyclic but not contractible.

Exercise 7. The Bockstein Homomorphism

Let ℓ be a prime and C a complex of free abelian groups. Write $\bar{C} = C/\ell C$, with the induced differential.

- (a) Prove that multiplication by ℓ and reduction modulo ℓ give a short exact sequence of complexes

$$0 \rightarrow C \xrightarrow{\ell} C \xrightarrow{\rho} \bar{C} \rightarrow 0.$$

Let $\delta_n : H_n(\bar{C}) \rightarrow H_{n-1}(C)$ be its connecting morphism. Define

$$\beta_n = H_{n-1}(\rho)\delta_n : H_n(\bar{C}) \rightarrow H_{n-1}(\bar{C}).$$

Describe β_n by lifting a cycle modulo ℓ and dividing its differential by ℓ . Prove that this description is independent of all choices.

- (b) Prove that $\beta_{n-1}\beta_n = 0$ and that β is natural with respect to chain maps between complexes of free abelian groups. Thus $(H_*(\bar{C}), \beta)$ is a complex of $\mathbb{Z}/\ell$ -vector spaces.
- (c) For every $r \geq 1$, consider

$$C : \quad 0 \rightarrow \mathbb{Z} \xrightarrow{\ell^r} \mathbb{Z} \rightarrow 0$$

in degrees 1, 0. Compute $H_*(C)$, $H_*(\bar{C})$, δ_1 , and β_1 . Explain exactly how $r = 1$ differs from $r > 1$.